Spelling Bee

This is a game in the NY Times Games portfolio.
The system generates 7 letters **without repetition:** a main letter, and six other letters.
These 7 letters are displayed with spaces between them, with the main letter written first.

The object of the game is for the player to guess as many legitimate words of at least 4 letters as possible that use only (a subset of) these letters allowing for letters to be used multiple times, such that the **main letter must appear** in every word. A score for each word guessed is computed, which (for our purposes) is based on the length of the word. A word that uses all 7 letters is called a PANGRAM and earns bonus points.

The system internally generates a list of all possible legitimate words using only these 7 letters and using the main letter. If the player guesses all these words that the system has computed, the player wins the "Queen Bee" accolade, and the game terminates. If the player quits before guessing all possible words the system has generated, the player's score is reported as well as the list of words that the player was unable to guess. The system continuously displays the words the player has guessed helpfully sorted alphabetically in three columns

The online game on NY Times also has a hint feature which reveals to the player how many words of each length it has in its list, as well as how many words in its list being with what 2-letter sequence. Here is an example.

Letters with main letter in bold written first.

C A I M O R T

WORDS: 63, POINTS: 297, PANGRAMS: 1

	4	**5**	**6**	**7**	**8**	**9**	**Σ**
A:	2	2	4	1	1	1	**11**
C:	11	12	4	-	2	-	**29**
M:	1	3	1	1	1	-	**7**
O:	1	-	-	-	-	-	**1**
R:	-	-	2	1	-	-	**3**
T:	2	3	4	3	-	-	**12**
Σ:	**17**	**20**	**15**	**6**	**4**	**1**	***63***

Two letter list:

AC-3 AO-1 AR-3 AT-4
CA-9 CI-4 CO-12 CR-4
MA-2 MI-3 MO-2
OR-1
RI-1 RO-2
TA-5 TO-3 TR-4

My guesses included:

AROMATIC (a pangram), CITRIC, CRITIC, ATOM, ATOMIC, ACAI, COMMIT, COMIC, MICRO, TACTIC, TACIT, ACTOR, AORTIC, ORCA, TRACT, TRACTOR, COMA, COMMA, TOCCATA, ATTRACT, ATTRACTOR, ROCOCO, MACRO, …

Your tasks:
- a function that generates 7 distinct letters of which one is designated the main letter;
- displaying the seven letters, with the main letter first;
- generating a list (used informally, you should select an appropriate data structure) of all dictionary words that use only these 7 letter and must include the main letter.
- User interface that takes in a guess from the player, checks that it uses only the relevant letters, includes the main letter, and checks if the guess is in the list of words generated earlier.
- Appropriate points can be awarded when a guess is in the target list of words (you can design your own scoring scheme).
- Displaying all words already guessed by the player.
- Appropriately friendly user interfaces and helpful error messages.
- Optionally displaying the hint grid and 2-letter list.

Learning outcomes expected:
- Correct choice of using mutable and immutable variables, use of constants; appropriate parameter passing — by reference or by copying. Please note ownership transfers.
- Familiarity with control flow — loops, conditionals, termination. Definite and indefinite iteration.
- **Function-level abstraction**
- Choice of **appropriate data structures** — arrays / vectors, indexing. Structs and Enums. The option type.
- Development of the program as **modules**: data encapsulation.
- Use of random word generators for the game, and for testing — particularly unit testing
- documentation of the code
- meta-documentation of the testing process.
- Using Cargo appropriately to build, check, test and release the project.

The system does not let me guess words not in the dictionary, and will not let me input words that use a letter outside the 7 given letters.